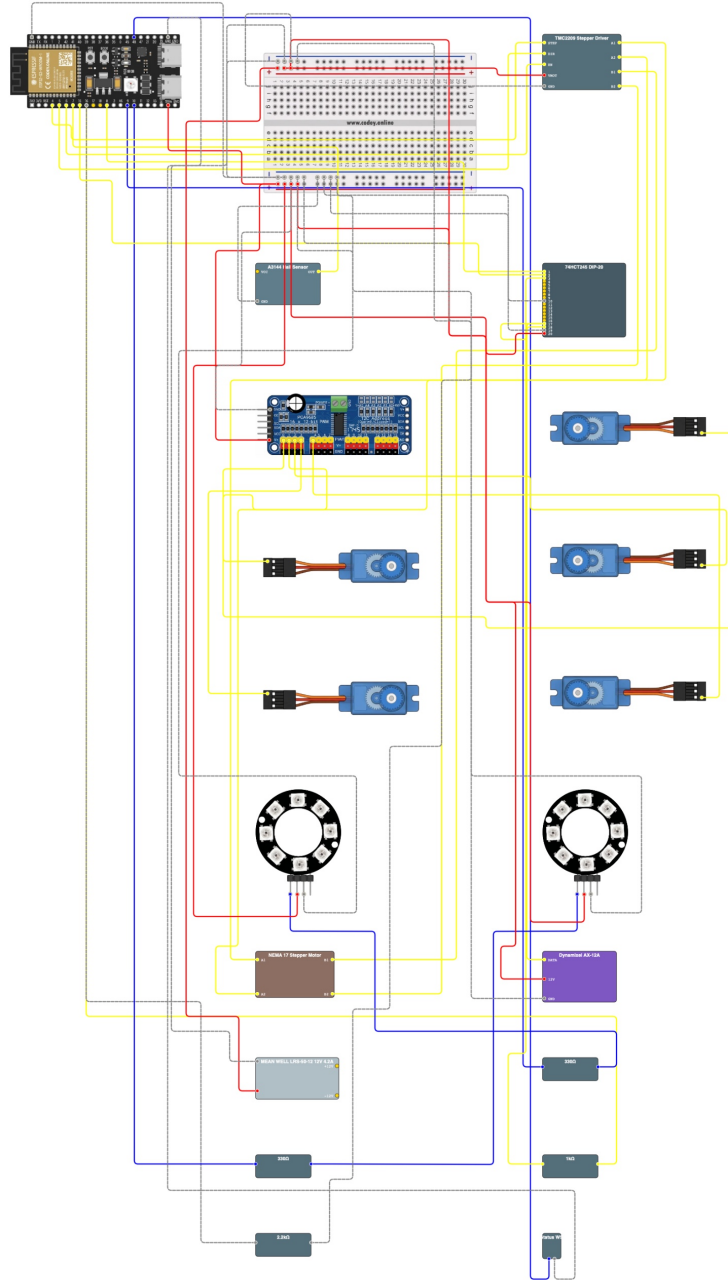




This diagram was generated with Codey.online.

Wiring connections

WIRE	FROM	TO
	ESP32-S3 N16R8 DevKitC 5Vin	Breadboard 400 pin +2.1
	ESP32-S3 N16R8 DevKitC GND	Breadboard 400 pin GND2.1
	MEAN WELL LRS-50-12 12V 4.2A +12V	Breadboard 400 pin +1.1
	MEAN WELL LRS-50-12 12V 4.2A -12V	Breadboard 400 pin GND1.1
	Breadboard 400 pin GND1.2	Breadboard 400 pin GND2.2
	Breadboard 400 pin +2.1	Pololu Mini Maestro 24ch V+
	Breadboard 400 pin GND2.3	Pololu Mini Maestro 24ch GND
	Breadboard 400 pin +2.2	WS2812 Rear LED Ring 16 LEDs 5V
	Breadboard 400 pin GND2.4	WS2812 Rear LED Ring 16 LEDs GND
	Breadboard 400 pin +2.3	WS2812 Front LED Ring 35 LEDs 5V
	Breadboard 400 pin GND2.5	WS2812 Front LED Ring 35 LEDs GND
	Breadboard 400 pin +1.2	TMC2209 Stepper Driver VMOT
	Breadboard 400 pin GND1.3	TMC2209 Stepper Driver GND
	Breadboard 400 pin +1.3	Dynamixel AX-12A 12V
	Breadboard 400 pin GND1.4	Dynamixel AX-12A GND
	Breadboard 400 pin GND2.6	A3144 Hall Sensor GND
	Breadboard 400 pin GND2.7	74HCT245 DIP-20 10
	Breadboard 400 pin GND2.8	74HCT245 DIP-20 19
	Breadboard 400 pin +2.4	74HCT245 DIP-20 20
	ESP32-S3 N16R8 DevKitC 4	TMC2209 Stepper Driver STEP
	ESP32-S3 N16R8 DevKitC 5	TMC2209 Stepper Driver DIR
	ESP32-S3 N16R8 DevKitC 6	TMC2209 Stepper Driver EN
	ESP32-S3 N16R8 DevKitC 7	A3144 Hall Sensor OUT
	ESP32-S3 N16R8 DevKitC 8	74HCT245 DIP-20 1
	ESP32-S3 N16R8 DevKitC 15	74HCT245 DIP-20 2
	74HCT245 DIP-20 3	1 k Ω 1
	1 k Ω 2	ESP32-S3 N16R8 DevKitC 16
	ESP32-S3 N16R8 DevKitC 16	2 . 2 k Ω 1
	2 . 2 k Ω 2	Breadboard 400 pin GND2.9
	ESP32-S3 N16R8 DevKitC 17	Pololu Mini Maestro 24ch RX
	ESP32-S3 N16R8 DevKitC 18	Pololu Mini Maestro 24ch TX
	ESP32-S3 N16R8 DevKitC 48	Onboard Status WS2812 LED ANODE +
	Onboard Status WS2812 LED CATHODE -	ESP32-S3 N16R8 DevKitC GND.1
	ESP32-S3 N16R8 DevKitC 9	3 3 0 Ω 1
	3 3 0 Ω 2	WS2812 Rear LED Ring 16 LEDs DI
	ESP32-S3 N16R8 DevKitC 10	3 3 0 Ω 1
	3 3 0 Ω 2	WS2812 Front LED Ring 35 LEDs DI
	74HCT245 DIP-20 17	74HCT245 DIP-20 18
	74HCT245 DIP-20 17	Dynamixel AX-12A DATA
	Pololu Mini Maestro 24ch CH0	MG996R Servo Lower Arm PWM
	Pololu Mini Maestro 24ch CH1	MG996R Servo Elbow PWM
	Pololu Mini Maestro 24ch CH2	MG90S Servo Neck Pan PWM

